

Indoor Plant Hardiness Score

Framework Writeup: Variables, Formula, and Scope

Bangalore East 560087 · 12th floor · May 2026

1. Variables Considered

Four variables were included in the final scoring model. Each met two criteria: (a) species-specific threshold data was available from cited published sources, and (b) the variable was directly measurable in the field without laboratory equipment.

1.1 Light Intensity (lux)

Light is the primary energy source for photosynthesis and the most commonly cited cause of indoor plant failure. Each species has a documented optimal lux range within which growth and flowering occur, and a survivable range outside which permanent damage begins. Data sourced from The Plant Runner, Deep Green Permaculture, and LeafPlantGarden — sources that provide numerical lux values rather than qualitative descriptors (low/medium/high). Measurable with a ₹300–500 lux meter or smartphone app.

1.2 Temperature (°C)

Temperature governs enzyme activity, respiration rate, and cellular integrity. Tropical houseplants — which constitute the majority of common Indian indoor species — typically thrive in a 18–27°C band but survive outside it. Optimal and survivable ranges differ per species. Sourced from PictureThis, Cafe Planta, IndoorPlantsWorld, and Bricks n Blooms. Measurable with any thermometer.

1.3 Humidity (% RH)

Relative humidity governs transpiration rate and affects stomatal function. Too low causes desiccation stress (brown leaf tips, wilting); too high promotes fungal growth. Species originating from tropical rainforest floors (peace lily, anthurium) have narrow high-humidity optimal bands; desert-origin species (ZZ, snake plant) have wide tolerant bands. Sourced from PLNTS.com, LeafPlantGarden, Thompson & Morgan. Measurable with a hygrometer (₹200–400).

1.4 Root Gap (cm)

Root restriction — the gap between pot diameter and root ball diameter — affects nutrient uptake, water retention, and growth rate independent of soil moisture. A rootbound plant (gap = 0 cm) experiences reduced growth, faster soil drying, and nutrient starvation. An oversized pot (gap > 5 cm) retains excess moisture and promotes root rot. General threshold sourced from Royal Horticultural Society repotting guidelines; slow growers (ZZ, snake plant) use a tighter optimal band of 2–3 cm, fast growers 4–5 cm. Measurable with a ruler at repotting.

2. Variables Not Considered

Six variables were identified as relevant but excluded from scoring for one of two reasons: no quantitative species-specific threshold data was found in published literature, or the variable requires laboratory measurement unavailable to a typical plant owner.

2.1 Air Pollution Tolerance (APT_I)

The Air Pollution Tolerance Index (APTI) is a validated composite metric derived from leaf chlorophyll content, ascorbic acid concentration, relative water content, and leaf pH. It directly quantifies a plant's ability to tolerate urban air pollutants (PM2.5, NO_x, SO₂). Excluded because: (a) no published APTI values were found for ZZ plant, snake plant, peace lily, or anthurium as indoor houseplants — APTI literature covers outdoor urban trees exclusively; (b) measurement requires spectrophotometry and laboratory conditions. Flagged for future inclusion if leaf biochemistry data becomes available.

2.2 Ventilation / Airflow (m/s)

Sustained airflow above 0.5 m/s has been documented to induce oxidative stress in some species (Kew Gardens study on Monstera and Calathea). Excluded because: (a) this threshold applies to two species only — no general or species-specific airflow threshold was found for ZZ, snake plant, peace lily, or anthurium; (b) empirical evidence from this study (anthurium surviving >1 year at estimated 1.0 m/s near a north-facing balcony) contradicts the cited threshold, suggesting the data is insufficient. Retained as a qualitative flag only.

2.3 Water TDS (ppm) and Soil pH

Both variables affect nutrient availability and osmotic stress. TDS and pH thresholds were identified per species from published sources (Rutgers NJAES for TDS; PictureThis, Thompson & Morgan for soil pH). Excluded from scoring because: the relevant measurement is soil TDS and soil pH — not water TDS or water pH. Water quality affects soil quality gradually over months of repeated watering. Without a soil TDS meter or pH probe, the scored variable cannot be measured by the plant owner. Water quality is retained as a qualitative modifier — the water source type (RO, borewell, tap) informs a flag, not a score.

2.4 Watering Frequency (days)

Initially considered as a scored factor. Excluded after identifying a structural problem: published watering intervals (e.g. 'every 21–56 days' for ZZ plant) are calibrated to temperate climates with lower ambient temperature, lower evapotranspiration, and standardised pot sizes. In Bangalore May conditions (27–32°C, 12th floor, small pots), the same species dries out in 7 days — consistent with the published interval being wrong for this context, not the owner's practice being wrong. Watering frequency is now treated as an output of the framework — computed from drought tolerance, pot size, temperature, humidity, and light — rather than a scored input. Drought tolerance in days (days plant survives after soil reaches dry) was also searched for all species; no quantitative published data was found.

2.5 Dust Accumulation, Pest Susceptibility, Fertiliser Sensitivity

Identified as potentially relevant urban factors. Excluded entirely: no quantitative threshold data exists in any accessible database for these variables at the species level. Noted as future scope.

3. Formula Used

3.1 Four-Value Threshold Structure

Each scored factor uses four values per species, not two. The distinction between survivable range and optimal range is critical — care guide data typically provides the survivable range; optimal ranges were sourced separately from species-specific studies.

surv_min	— below this, permanent damage begins
opt_min	— below this, plant survives but does not thrive
opt_max	— above this, plant survives but does not thrive
surv_max	— above this, permanent damage begins

3.2 Shelford Approximation (Four-Value)

The scoring function is based on Shelford's Law of Tolerance — a bell-shaped response curve where performance peaks at the ecological optimum and decays toward zero at survivable limits. Validated in plant biology literature (Antenozio et al., 2024; Wang-Engel cardinal temperature model). The four-value implementation gives a flat score of 5 within the optimal band and decays asymmetrically on each side.

```
score = 5                                     if opt_min ≤ actual ≤ opt_max
score = 5 × exp(-0.5 × ((actual - opt_min) / σL)²)  if actual < opt_min
score = 5 × exp(-0.5 × ((actual - opt_max) / σR)²)  if actual > opt_max

σL = (opt_min - surv_min) / 3
σR = (surv_max - opt_max) / 3
```

3.3 Normalised Sensitivity Weights

Factor weights are derived from optimal band width, normalised to a 0–1 scale using physical limits of each variable. This prevents unit scale differences (lux vs °C vs % vs cm) from distorting weights — the primary flaw of using raw tolerance widths.

```
Physical limits used for normalisation:
light: 0–10,000 lux · temperature: 0–50°C
humidity: 0–100% · root gap: 0–10 cm

norm_opt_width = (opt_max - opt_min) / (phys_max - phys_min)
sensitivity     = 1 / norm_opt_width
weight         = sensitivity / Σ sensitivities [sums to 1.0 per plant]
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3.4 Final Score

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final_score = Σ [score(factor) × weight(factor)] × 2 → 0–10 scale
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3.5 Score Interpretation

Score range	Label	Meaning
8–10	Perfect	All factors at or near optimal
6–8	Good	Minor deviations from optimal, no stress
4–6	Average	Noticeable deviation, plant survives but not thriving
2–4	Risky	Significant deviation, long-term stress likely
0–2	Fatal	Conditions near or beyond survivable limits

4. Scope of Improvement

4.1 Expand to optimal threshold data

The most impactful improvement is sourcing four-value threshold data (surv_min, opt_min, opt_max, surv_max) for all 30 species across all factors from controlled experiments. The TRY Plant Trait Database (try-db.org) is the most promising source for temperature cardinal values; no equivalent exists for light and humidity at species level. A scraping

and data pipeline targeting NC State Extension, Deep Green Permaculture, and Houseplants Nook is the recommended first step.

4.2 Asymmetric stress modelling

The Shelford approximation as implemented is symmetric within each side of the optimal band (σ_L and σ_R are independent, but the curve shape is identical). In reality, overwatering kills faster than underwatering for arid-origin species; high TDS is more damaging than low TDS. A skewed distribution (e.g. log-normal) per factor per species would better reflect empirical stress response, but requires controlled stress-response data not currently available in literature.

4.3 Acclimation factor

The current framework scores environment vs. optimal thresholds — it does not account for plant acclimation over time. A plant that has survived suboptimal conditions for 12+ months has physiologically adjusted its optimal range. An acclimation modifier — user-input duration in current conditions — would shift the effective optimal band toward actual conditions proportionally. The relationship between acclimation duration and threshold shift is not documented in literature for these species; this remains qualitative until controlled acclimation studies are conducted.

4.4 APTI inclusion

Air Pollution Tolerance Index inclusion requires leaf biochemistry measurement (chlorophyll, ascorbic acid, RWC, pH) per species per growing condition. As urban air quality data for Bangalore East becomes more granular (CPCB monitoring stations), APTI can be estimated from species-level published data where it exists. Priority species for APTI inclusion from the 30-plant dataset: Areca palm, Money plant, Spider plant — all have published APTI values from Indian urban studies.

4.5 Soil measurement integration

TDS and pH are correctly modelled as soil variables, not water variables. Affordable soil EC/TDS meters (₹500–1,500) and soil pH meters (₹400–1,200) are commercially available. If the owner measures soil TDS and pH directly, both variables can be re-introduced as scored factors with species-specific thresholds from published sources. This would add two high-quality scored dimensions to the model with minimal framework changes.

4.6 Watering frequency as output

The framework currently excludes watering frequency as a scored input. The next step is to implement it as a computed output — a recommendation to the plant owner derived from drought tolerance, pot size, temperature, humidity, and light intensity. This requires drought tolerance data (days plant survives after soil reaches dry) which was not found in literature for these species. Community-sourced data from platforms like Greg.app or Planta may be the only available source at this time, with the caveat that such data is climate-uncontrolled.

4.7 30-plant dataset and scoring pipeline

The current framework has been applied to four plants manually. A scraping pipeline targeting the identified sources (NC State Extension, Houseplants Nook, Deep Green Permaculture, The Plant Runner) would build the full 30-plant threshold dataset. A Python script using BeautifulSoup and requests is the recommended approach, with SQLite as the storage backend and computed sensitivity/weight columns derived at query time rather than stored statically.